Pediatric Nephrology (2021) 36:2453–2461
https://doi.org/10.1007/s00467-021-04946-8

ORIGINAL ARTICLE

Predicting acute rejection in children, adolescents, and young adults with a kidney transplant by assessing barriers to taking medication

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Received: 8 September 2020 / Revised: 15 December 2020 / Accepted: 12 January 2021 / Published online: 27 January 2021
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Abstract

Background Nonadherence to immunosuppression is common among pediatric, adolescent, and young adult kidney transplant recipients and a leading cause of graft loss. Assessing barriers to medication adherence in clinical practice may identify patients at risk for rejection and provide therapeutic targets.

Methods Kidney transplant patients and/or their caregivers were assessed for 14 barriers to medication adherence using the barriers assessment tool. We compared rejection rates between patients with at least one reported adherence barrier to those without reported adherence barriers using a Kaplan-Meier estimator and Cox proportional hazard models to adjust for other mediators of acute rejection at 2 years following barriers assessment.

Results Ninety-eight patients were assessed for barriers to adherence. Over the 2-year observation period, 22 patients developed biopsy-proven acute rejection (BPAR). Kaplan-Meier estimates show that patients with an identified barrier to adherence were more likely to have BPAR ($p = 0.02$) than patients without an identified barrier in the 24 months following barriers assessment. The median time to rejection for patients who experienced acute rejection was 175.5 days (IQR 63–276 days) from the time of barriers assessment. An identified barrier to adherence remained the only statistically significant predictor of BPAR with Cox modeling (HR 2.6, $p = 0.04$), after accounting for age, sex, and race.

Conclusions Pediatric and adolescent kidney transplant recipients with identified adherence barriers are at increased risk for acute rejection. Barriers to adherence provide a potentially modifiable therapeutic target that can be assessed in clinic to guide targeted interventions.

Keywords Medication adherence · Immunosuppression adherence · Pediatric transplantation · Acute rejection · Pediatric nephrology

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Abbreviations

AMR	Antibody-mediated rejection
BAT	Barriers assessment tool
BPAR	Biopsy-proven acute rejection
CAKUT	Congenital anomaly of the kidney and urinary tract
DSA	Donor-specific antibody
CKD 5	Stage 5 chronic kidney disease
TCMR	T cell–mediated rejection

Introduction

Advancements in surgical techniques and immunosuppressive regimens have improved allograft and patient survival following kidney transplantation; however, the gains have

largely been limited to the first year post-transplant [1]. While the issues related to long-term allograft survival are complex, suboptimal adherence to the immunosuppression regimen following kidney transplantation is common—particularly in adolescents and young adults—and leads to increased incidence of acute rejection [2], a major cause of allograft failure [1]. Reducing acute rejection episodes has been shown to decrease chronic rejection and improve allograft survival [1]. This leads to increased life expectancy [3] and quality of life [4], with significantly lower health care costs [3] compared to dialysis. Therefore, medication adherence is a potentially modifiable risk factor for interventions to improve long-term graft survival in post-transplant patients. Unfortunately, physicians have limited training and ability to accurately identify nonadherence [5, 6] and associated barriers to taking immunosuppressive medications despite physician recognition that nonadherence is both common and an important factor in clinical outcomes [7, 8].

Adolescence is a particularly vulnerable period for nonadherence due to less parental supervision, feelings of invincibility, lack of fully developed executive functioning skills, peer pressure, and increased vulnerability to risk taking [9–11]. Children, adolescent, and young adult patients and their caregivers often experience barriers to taking/administering their immunosuppressive medication, including disliking the taste, forgetting, difficulty swallowing pills, and competing demands [2, 12]. Identifying these barriers is a necessary first step for any effort to improve adherence behaviors [13]. In the research setting, barriers to taking medication have been linked to adherence across multiple chronic conditions, including solid organ transplant, diabetes, and epilepsy [10, 14–16]. In solid organ transplant recipients, specific adherence barriers endorsed by *parents* have been associated with missed doses or erratic drug levels, whereas specific adherence barriers endorsed by *patients* are associated with rejection episodes and/or patient death [16]. In addition, adherence barriers are more strongly linked to particular outcomes in specific age groups (e.g., adolescents who disliked medication taste had lower medication adherence rates) [10].

While studies have demonstrated the value of screening for adherence barriers in the research setting, there are few studies examining implementation of adherence barrier screening in clinical care [12]. Our kidney transplant program standardized and integrated screening for adherence barriers as part of standard post-transplant care using the barriers assessment tool (BAT), an easily administered tool (< 1 min) for use in the clinic setting [12]. The BAT was developed based on existing validated research tools, such as the Parent Medication Barriers Scale (PMBS)/Adolescent Medication Barriers Scale (AMBS) [17], Cystic Fibrosis and Asthma Barriers to Adherence Interview [15], and Pediatric Epilepsy Medication Self-Management Questionnaire [18]. While these existing tools are well suited and proven in the research setting and occasional clinical use, our busy clinical setting and lack of additional resources to dedicate to administering a barriers tool in clinic required the creation of a new tool that could be easily understood and completed by patients and/or their caregivers. Once completed, the tool also needed to be easily scored and interpreted by the clinician as part of a routine clinic visit. To this end, we used a checklist format (i.e., yes/no) with a limited number of known adherence barriers that can be easily reviewed and interpreted in the clinic by any provider [12]. The aim of this study was to determine whether adherence barriers identified using this tool would predict rejection in the clinical setting after accounting for other known risk factors for rejection. We hypothesized that patients who endorsed at least one adherence barrier via either patient or caregiver report would be more likely to experience biopsy-proven acute rejection (BPAR) compared to patients with no reported adherence barriers.

Methods

We performed a prospective cohort study to compare incidence of acute rejection in patients who identified a barrier to taking their immunosuppressive medications and those who did not. We followed each patient for 24 months after completing the BAT or until they experienced acute rejection. This study was approved by the site's Institutional Review Board.

Assessment of adherence barriers

Our group previously described the development of the BAT and our process of reliably administering it as standard-of-care for all patients in our kidney transplant program [12]. The BAT is a 14-item checklist that asks "what gets in the way of taking your immunosuppression" and elicits responses that correspond to barriers most commonly identified in the pediatric adherence literature related to logistical issues (e.g., forgetting, inconvenience), ingestion difficulties (e.g., swallowing, taste), efficacy (e.g., feel I don't need it), financial difficulties, regimen characteristics (e.g., too many medications, side effects), and patient-specific issues (e.g., refusal by child, embarrassment) [12]. It is completed during clinic intake in under 60 s. For patients ≥ 10 years old, both the patient and their caregiver (if present) are asked to complete the BAT. If the patient was < 10 years old, only the caregiver is asked to complete the BAT. All kidney transplant patients seen across six outpatient clinic locations are eligible to complete the BAT. In clinical practice, patients complete the BAT every 6 months. For this study, the first BAT completed by a patient and/or their caregiver from January 2015 through August 2016 was used for analysis. During the study period,

there was no formal or standardized approach to addressing adherence barriers that were identified; however, providers had full access to the results of the BAT and addressed any identified adherence barriers as they deemed appropriate.

Diagnosis of acute rejection and biopsies

All patients for whom a BAT was completed were followed and assessed for the development of BPAR from the time of BAT assessment for 24 months. The time of BAT assessment was chosen as the starting point for rejection risk assessment because we hypothesized that the information gathered from barriers assessment is relevant to subsequent rejection risk. At our center, biopsies for kidney transplant patients are "for cause" and indicated when there is kidney dysfunction (e.g., elevated creatinine), de novo donor-specific antibodies (DSAs), new proteinuria, and/or new hypertension. The presence or absence of a barrier to adherence did not factor into the decision to perform a kidney biopsy. Each biopsy was read by one of two kidney transplant pathologists who were blinded to the study and scored according to Banff classification. The presence or absence of biopsy-proven acute cellular (≥ Banff 1a) [19] or antibody-mediated rejection [20] during the observation period was the primary outcome measure. Since our center does not perform protocol biopsies, we sought to evaluate for ascertainment bias by analyzing the overall number of biopsies in the population during the observation period, the percentage of biopsies performed in those who did and did not endorse an adherence barrier, and the percentage of biopsies that showed rejection between patients who did and did not endorse an adherence barrier.

Procedure

All kidney transplant patients who had BAT data (self- and/or caregiver-report) between January 2015 and August 2016 were included in the study. If either the patient and/or the caregiver identified a barrier, the patient was noted to have an adherence barrier. There were no exclusion criteria. For patients over the age of 10 years who had developmental delay, only their caregiver completed the BAT. All patients cared for in our kidney transplant clinic were asked to compete the BAT during this time as part of routine clinical care. Electronic health records were reviewed to identify baseline demographic (age, sex, race) and medical information/markers of immunologic status (cause of kidney disease, calculated panel reactive antibodies (cPRA) prior to transplant, type of transplant, transplant episode, HLA mismatch, induction immunosuppression, presence of delayed graft function (DGF) post-transplant, primary immunosuppression, prior rejection episodes, the presence of HLA DSAs prior to BAT assessment, development of de novo HLA DSAs after BAT assessment, biopsy rates, and BPAR) for the 24-month period following the BAT assessment. DSAs were assessed using the Luminex® microbead platform prior to transplant and at 3, 6, and 12 months post-transplant and annually thereafter. A pre-transplant MFI cutoff of 1500 and post-transplant MFI cutoff of 1000 are considered positive. Due to the sample size and uncommon outcome event of acute rejection, we did not analyze whether the number of barriers for a given patient or specific barriers was linked to acute rejection.

Statistical analysis

Descriptive analysis was performed to describe the distribution of the variables of interest. Two-sample *t* tests or Wilcoxon rank sum tests were used to compare continuous variables of patients who identified an adherence barrier to those that did not. Categorical variables were compared using chi-squared test or Fisher's exact test. The Kaplan-Meier estimate was used to illustrate the difference in survival curves of time to rejection between two groups. Cox proportional hazard models were used to evaluate the correlation between time to rejection and an identified adherence barrier adjusted for key variables a priori selected based on reported risk factors in the literature known to affect rejection (i.e., age, sex, race) [21–27], as well as by other patient characteristics. Model selection was based on stepwise criterion specifying that age, sex, and race were included in the Cox model. All analyses were performed using SAS version 9.4. A p value of less than 0.05 was considered statistically significant.

Results

Patient cohort and characteristics

From January 2015 to August 2016, 61 patients and 80 caregivers completed the BAT, representing 98 unique patients (Table 1), which represents 88% of 111 possible patients in our care during that period. The 13 patients not included in this study did not receive the BAT during the study period. To evaluate for ascertainment bias by not including these patients, their demographic data was compared to that of the 98 patients included in this study. The date of January 1, 2015, was used to calculate age and time since transplant and to determine prior DSA status. There was no difference between groups with regard to age, sex, race, stage 5 chronic kidney disease (CKD 5) cause, DGF, transplant type, transplant episodes, pre-transplant cPRA, induction immunosuppression, HLA mismatch, primary immunosuppression, prior rejection, or prior DSA. The 13 unenrolled patients did have their transplants longer, with mean time since transplant of 2835 days (± 1584 days ($p < 0.01$)). Eighty-six of the 98 patients (88%) remained in this cohort for the full 2-year observation period. Eleven patients transitioned to adult care and

Table 1 Patient characteristics

Patient characteristic	Total cohort, *n* = 98	Barrier identified, *n* = 42	No barrier identified, *n* = 56
Age in years			
Mean (SD)	13.6 (6.5)	13.7 (6.1)	13.5 (6.8)
Sex			
Male	52 (53%)	22 (53%)	30 (54%)
Race			
White	67 (68%)	28 (67%)	39 (69%)
Black	19 (20%)	9 (22%)	10 (18%)
Biracial	6 (6%)	3 (7%)	3 (5%)
Asian	3 (3%)	1 (2%)	2 (4%)
Hispanic	3 (3%)	1 (2%)	2 (4%)
CKD 5 cause			
CAKUT	47 (48%)	14 (33%)	33 (59%)
Genetic	22 (23%)	15 (36%)	7 (13%)
Glomerular	11 (11%)	4 (10%)	7 (13%)
Medication	5 (5%)	1 (2%)	4 (7%)
Vascular disease	4 (4%)	2 (5%)	2 (4%)
Oncologic	2 (2%)	2 (5%)	0 (0%)
Other	7 (7%)	4 (10%)	3 (5%)
Type of transplant			
Living	72 (73%)	28 (67%)	44 (79%)
Deceased	26 (27%)	14 (33%)	12 (21%)
Transplant episode			
First	93 (95%)	40 (95%)	53 (94%)
Second	4 (4%)	2 (5%)	2 (4%)
Third	1 (1%)	0 (0%)	1 (2%)
Primary IS			
Tacrolimus[a]	77 (79%)	32 (77%)	45 (80%)
Sirolimus	16 (16%)	8 (19%)	8 (14%)
Cyclosporine	2 (2%)	1 (2%)	1 (2%)
Everolimus	1 (1%)	0 (0%)	1 (2%)
Prednisone	1 (1%)	0 (0%)	1 (2%)
Belatacept	1 (1%)	1 (2%)	0 (0%)
Prior rejection?			
Yes	26 (27%)	13 (31%)	13 (23%)
DSAs			
Prior to BAT	15 (15%)	8 (19%)	7 (13%)
De novo after BAT[b]	12 (13%)	6 (14%)	6 (11%)
Any biopsy?			
Yes	46 (47%)	24 (57%)	22 (39%)
Time since Transplant (days)	1422 (1589)	1148 (1338)	1627 (1738)
DGF			
Yes	3 (3%)	1 (2%)	2 (4%)
Induction			
Basiliximab	80 (88%)	37 (92%)	43 (84%)
Thymoglobulin	7 (8%)	3 (8%)	4 (8%)
None	4 (4%)	0 (0%)	4 (8%)
PRA			
0–20%	96 (98%)	42 (100%)	54 (96%)
21–80%	1 (1%)	0 (0%)	1 (2%)

Table 1 (continued)

Patient characteristic	Total cohort, $n = 98$	Barrier identified, $n = 42$	No barrier identified, $n = 56$
> 80%	1 (1%)	0 (0%)	1 (2%)
HLA mismatch			
0/6	4 (4%)	2 (5%)	2 (4%)
1/6–5/6	78 (80%)	31 (74%)	47 (85%)
6/6	15 (16%)	9 (21%)	6 (11%)

SD standard deviation, *CAKUT* congenital anomaly of the kidney and urinary tract, *IS* immunosuppressant, *CKD 5* stage 5 chronic kidney disease, *DSA* donor-specific antibodies, *BAT* barriers assessment tool, *DGF* delayed graft function, *PRA* panel reactive antibody, *HLA* human leukocyte antigen

[a] Four patients were taking tacrolimus in combination with everolimus; two patients were taking tacrolimus in combination with sirolimus—they were included in the tacrolimus group for primary immunosuppression

[b] $n = 95$ (three patients without DSA checked after completion of barriers assessment tool)

one patient died. The median observation time for the 12 patients who left the cohort was 641 days (1.7 years).

Endorsement of barriers to adherence

Twenty-six (43%) patients and 28 (34%) caregivers identified at least one barrier to adherence, representing 42 unique patients (43% of the cohort). No significant differences were noted between those who did or did not endorse adherence barriers with respect to age, sex, race, transplant type (living vs. deceased), whether the patient had a prior transplant, prior rejection episode, and pre-existing or de novo DSA. However, the most common cause of CKD 5 among those with an adherence barrier was genetic (15/42, 36%), whereas congenital anomalies of the kidney and urinary tract (CAKUT) was most common among those without an adherence barrier (33/56, 59%) ($p = 0.02$).

Analysis of rejection risk

Twenty-two patients (22%) experienced acute cellular rejection (TCMR) and/or antibody-mediated rejection (AMR) during the follow-up period, while 76 patients (78%) did not have any acute rejection. Eighteen patients had a single diagnosis of TCMR: 12 patients with Banff 1A, 5 patients with Banff 1B, and 1 patient with Banff 2A, while 2 patients had a single diagnosis of acute AMR. Two patients had mixed rejection, 1 patient with Banff 1A TCMR and acute AMR and 1 patient with Banff 1B TCMR and acute AMR. Kaplan-Meier survival curve (Fig. 1) displays the "rejection free probability" for patients with an identified barrier and patients without an identified barrier in the 24 months after BAT assessment. The patients with an identified adherence barrier were more likely to experience acute rejection than patients without an identified barrier ($p = 0.02$). The median time to rejection for patients who experienced acute rejection was 175.5 days (IQR 63–276 days) from the time of barriers assessment. Cox modeling was performed to account for other known risk factors for acute rejection. After model selection including age, sex, and race, the only significant covariate was an identified barrier to adherence. In this analysis, patients with an adherence barrier were 2.6 times more likely to develop subsequent BPAR in the 24 months after BAT assessment compared to patients without an adherence barrier ($p = 0.04$) (Table 2).

Biopsies

Forty-six patients (47%) in our sample received a kidney biopsy. For patients with an adherence barrier, 24 (57%) were biopsied compared to 22 (39%) of patients without an adherence barrier ($p = 0.08$). When evaluating the percentage of biopsies with rejection in those who did and did not endorse an adherence barrier, we did not find a difference between the two groups. In this sub-analysis, 14 of 24 biopsies in patients (58%) with an adherence barrier had acute rejection, while 8 of 22 biopsies in patients (36%) without an adherence barrier had acute rejection ($p = 0.14$).

Table 2 Cox proportional hazard model for rejection in the 24 months following BAT assessment

Risk of biopsy-proven acute rejection after BAT assessment

	Hazard ratio	95% confidence interval	*p* value
Identified barrier	2.61	1.03–6.57	0.04
Age	0.98	0.91–1.05	0.60
Female	1.91	0.77–4.75	0.17
White race[a]	1.53	0.55–4.26	0.42

[a] Comparison is White vs. non-White

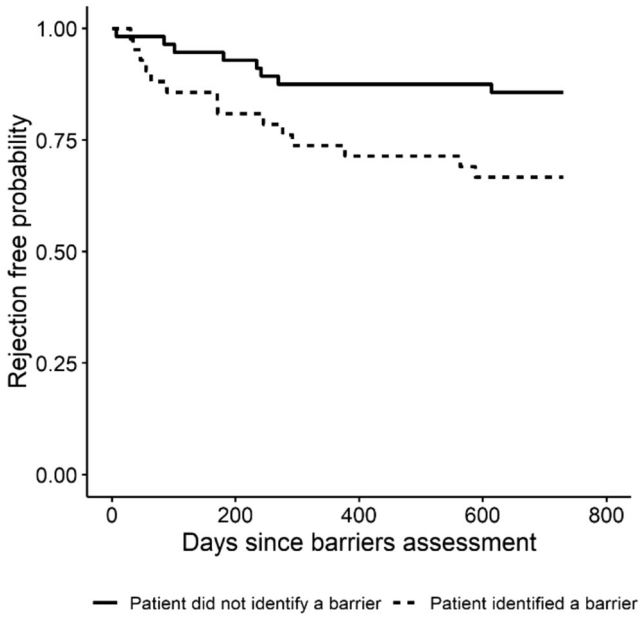


Fig. 1 Kaplan-Meier curve showing patients with an identified barrier compared to patients without an identified barrier in the 24 months after BAT assessment with regard to subsequent biopsy-proven acute rejection, $p = 0.02$

Donor-specific antibodies

Ninety-five of the 98 patients (97%) had DSA checked before and after completing the BAT. Fifteen patients had DSA prior to completing the BAT and all 15 of these patients continued to have DSA throughout the observation period. One of these patients had only class I DSA, one had class I and II DSA, and 13 had only class II DSA. Twelve of 95 patients (13%) developed de novo DSA during the observation period. All 12 patients developed class II DSA, with 3 of them also developing class I DSA. Of the 42 patients with an adherence barrier, 8 (19%) had pre-formed DSA while 6 (14%) developed de novo DSA during the observation period. For the patients without an adherence barrier, 7 (13%) had pre-formed DSA while 6 (11%) developed de novo DSAs during the observation period. There was no significant difference between the number of patients with pre-formed DSAs or the development of de novo DSA between patients with and without an adherence barrier ($p = 0.37$, 0.67, respectively).

Discussion

In this study, patients with an identified adherence barrier had a 2.6-fold increased risk of subsequently developing BPAR in the following 24 months. This finding remained after adjusting for other known factors known to be associated with acute rejection, such as age, sex, and race. Our sub-analysis of rejection rates suggests an equal number of biopsies being performed in those with and without barriers, suggesting there was no ascertainment bias (e.g., finding rejection in the group with barriers because more biopsies were performed); however, we acknowledge that bias cannot be completely eliminated and have made an earnest effort to minimize it in this analysis. Overall, these data suggest that patients who have a barrier to taking their medication identified by themselves or their caregiver are at increased risk for rejection. A barrier to adherence was common in this cohort with almost half of caregivers and a third of patients identifying a barrier to adherence. Barriers to adherence have been widely reported as being at least as common or more common than our finding across other pediatric chronic conditions [9, 28–30] Fortunately, unlike most rejection risk factors, the assessment of adherence barriers provides actionable data to the clinical team with regard to why a patient may be exhibiting suboptimal adherence and what might be done to mitigate the risk of rejection. Given their prevalence, routine assessment and monitoring of adherence barriers is indicated, especially for those who have endorsed a barrier. We must balance routine assessment with survey fatigue in clinical practice and as such, we chose to assess every 6 months. Using clinical judgment, there may be times when the BAT should be administered more frequently, such as a new psychosocial stressor or significant change in family dynamic. At the time of this study, there were no standardized responses or interventions for any given barrier and the clinician could react to identified barriers at their discretion. It is possible that this could attenuate any signal for subsequent rejection risk; however, the fact that we still see an increased risk for rejection when an adherence barrier is identified is significant. Interestingly, patients with CAKUT were half as likely to identify a barrier in this cohort compared to other diagnoses. One possible reason for this is CAKUT tends to be evident at birth, which means these families have likely adjusted to the illness for a longer period of time and potentially addressed barriers experienced early on in the disease process.

To date, there are few models in pediatric kidney transplantation that identify patients at increased risk for acute rejection [27, 31]. Prior studies utilize risk factors that are either non-modifiable for patients and clinicians (e.g., race, sex) or difficult to prevent and treat (e.g., DGF, prior sensitization). Additionally, prior approaches have included measures of adherence based on provider perception, self-report of missed doses, or biomarkers of immunosuppression adherence (e.g., coefficient of variation for immunosuppression drug levels) [27, 31]. Unfortunately, while these markers may be useful at identifying patients at increased risk for acute rejection and decreased graft survival, they provide little insight or opportunity for providers to implement interventions to prevent poor outcomes and have not been widely integrated into clinical practice post-transplant. Furthermore, adherence measurement that focuses on drug levels can be affected by several variables, including fluid shifts in the early post-transplant

period, interacting medications, pharmacokinetic factors affecting drug absorption or metabolism, and clinician drug dosing behaviors [32]. Lastly, provider and self-report of adherence are often erroneous or inflated [5, 6, 33]. In contrast, patient- or caregiver-identified barriers to adherence represent a specific therapeutic target that may be amenable to interventions. For example, if a patient endorses disliking the taste of their medication, the healthcare provider can change medications or work with pharmacy to mask the flavor and potentially improve medication adherence. Assessing for adherence barriers for solid organ transplant patients in the research literature is not a novel finding. The PMBS and AMBS are validated instruments that have been reported extensively as part of research studies to identify adolescent and parental barriers to adherence [16, 17, 34–36]. What is novel is the finding that the systematic assessment of barriers to adherence using the BAT—a simple tool that can be used as standard of care in clinical practice—was able to identify patients at higher risk for rejection. Our study findings show that, when applied systematically during routine clinical care, the BAT provides clinicians a "behavioral biomarker" that is easily identified, empowering clinicians with the information needed to partner with their patients to address challenges and behaviors patients and their caregivers are experiencing.

The TAKE-IT study [13], a randomized clinical trial to improve adherence in pediatric kidney transplantation, supports the idea of assessing and addressing adherence barriers. In this seminal clinical trial, identified adherence barriers were addressed via "action-focused problem solving." Patients who received the intervention received reminders, coaching, and problem solving around adherence barriers and demonstrated improved adherence and timing compared to controls (e.g., social support and active listening). Our study adds to this intervention literature and compliments TAKE-IT results. Specifically, we found that adherence barriers increase the risk of subsequent rejection while the TAKE-IT study demonstrates the benefits of systematic and multimodal adherence interventions to address such identified barriers.

Our results should be interpreted in light of several limitations. First, this is a single-center study that may not be generalizable to other centers. Given our sample size, we could not conduct analyses to identify whether the number of barriers per patient affected the risk for rejection, or whether particular barriers are more common for certain individuals or more predictive of acute rejection. Likewise, we could not determine whether caregiver-identified vs. patient-identified barriers to adherence presented any difference in subsequent rejection risk, although this has been described in other patient populations [10]. An additional limitation is that we did not measure patient medication adherence directly, so while we hypothesize that the causal pathway of endorsing an adherence barrier leading to increased risk for subsequent rejection is due to suboptimal medication adherence, we do not have evidence to support that. However, prior literature would support these important relationships [16]. Clinicians were not blinded to the results of the barriers assessment in the clinic and were able to address identified barriers at their discretion, so our findings may be an underestimate of the true risk of rejection in those who identify a barrier to taking their medication without intervention. We did not perform protocol biopsies, so our study did not include subclinical rejection as an outcome. Lastly, the BAT has not been comprehensively validated. Future studies are needed to more fully evaluate the psychometric properties of the BAT. Given the findings of this study, we are in the process of conducting a comprehensive validation of the BAT. The BAT is also currently being utilized at other pediatric kidney transplant programs, so we hope to use multicenter data to assess generalizability.

Overall, our findings indicate that patients who have an adherence barrier identified by themselves or a caregiver are 2.6 times as likely to experience subsequent BPAR in the 24 months following BAT assessment. Notably, this risk persists after controlling for other known risk factors for acute rejection. This feasible assessment of adherence barriers provides valuable information and is a necessary first step toward developing and implementing interventions to reduce the burden of acute rejection. Patients and clinicians can then partner in overcoming identified barriers using shared decision-making and other approaches shown to be effective [13, 37, 38]. Since this study was completed such interventions have been designed and are being used in our transplant clinic. These interventions include shared decision-making tools that were created for the five most commonly identified barriers to adherence that are reviewed by the patient and the healthcare team prior to the patient leaving clinic. Additional interventions include referrals to social work for logistical and financial barriers and referral to psychology for behavioral barriers (e.g., child oppositional behaviors), as well as focused discussions with the healthcare provider for barriers related to confusion or worries about the treatment regimen. Next steps will be to describe the effect these interventions have on our clinical outcome of interest, acute rejection. Additional future studies should be conducted to expand barriers assessment to other centers in order to identify which barriers are most important for various clinical outcomes, including medication nonadherence, acute and chronic rejection, development of DSA, decline in glomerular filtration rate, and graft failure. Such work is underway in the Improving Renal Outcomes Collaborative [39].

Authors' contributions Participated in research design: CDV, KLR, ALHP, ACM, DKH

Participated in writing of the paper: CDV, KLR, ACM, DKH

Participated in performance of the research: CDV, KLR, ALHP, ACM, DKH

Participated in data analysis: CDV, BZ, ACC, ACM, DKH

Funding CDV received funding from the National Center for Advancing Translational Sciences of the National Institutes of Health, under Award Number 2KL2TR001426-05A1.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Disclaimer The content is solely the responsibility of the authors and does not necessarily represent the official views of the NIH.

References

1. Tejani A, Ho PL, Emmett L, Stablein DM, North American Pediatric Renal Transplant Cooperative Study (NAPRTCS) (2002) Reduction in acute rejections decreases chronic rejection graft failure in children: a report of the North American Pediatric Renal Transplant Cooperative Study (NAPRTCS). Am J Transplant 2:142–147
2. Dobbels F, Ruppar T, De Geest S, Decorte A, Van Damme-Lombaerts R, Fine RN (2010) Adherence to the immunosuppressive regimen in pediatric kidney transplant recipients: a systematic review. Pediatr Transplant 14:603–613
3. 2016 USRDS annual data report: Epidemiology of kidney disease in the United States, in USRDS Annual Data Report, N.I.D.D.K. National Institutes of Health, Editor. 2016, United States Renal Data System: Bethesda, MD.
4. Goldstein SL, Rosburg NM, Warady BA, Seikaly M, McDonald R, Limbers C, Varni JW (2009) Pediatric end stage renal disease health-related quality of life differs by modality: a PedsQL ESRD analysis. Pediatr Nephrol 24:1553–1560
5. Nevins TE, Nickerson PW, Dew MA (2017) Understanding medication nonadherence after kidney transplant. J Am Soc Nephrol 28: 2290–2301
6. Osterberg L, Blaschke T (2005) Adherence to medication. N Engl J Med 353:487–497
7. Mehta P, Steinberg EA, Kelly SL, Buchanan C, Rawlinson AR (2017) Medication adherence among adolescent solid-organ transplant recipients: a survey of healthcare providers. Pediatr Transplant 21(7). https://doi.org/10.1111/petr.13018
8. Skeens MA, Gerhardt CA, Bajwa R, Akard TF (2020) Toward a better understanding: an exploration of provider perceptions in pediatric hematopoietic stem cell transplant adherence. Pediatr Transplant 24:e13786
9. Smith AW, Mara CA, Modi AC (2018) Adherence to antiepileptic drugs in adolescents with epilepsy. Epilepsy Behav 80:307–311
10. Gutierrez-Colina AM, Smith AW, Mara CA, Modi AC (2018) Adherence barriers in pediatric epilepsy: From toddlers to young adults. Epilepsy Behav 80:229–234
11. Smith AW, Mara C, Ollier S, Combs A, Modi AC (2017) Rebellious behaviors in adolescents with epilepsy. J Pediatr Psychol 43:52–60
12. Varnell CD Jr, Rich KL, Nichols M, Dahale D, Goebel JW, Pai ALH, Hooper DK, Modi AC (2017) Assessing barriers to adherence in routine clinical care for pediatric kidney transplant patients. Pediatr Transplant 21. https://doi.org/10.1111/petr.13027
13. Foster BJ, Pai ALH, Zelikovsky N, Amaral S, Bell L, Dharnidharka VR, Hebert D, Holly C, Knauper B, Matsell D, Phan V, Rogers R, Smith JM, Zhao H, Furth SL (2018) A randomized trial of a multicomponent intervention to promote medication adherence: the Teen Adherence in Kidney Transplant Effectiveness of Intervention Trial (TAKE-IT). Am J Kidney Dis 72:30–41
14. Gray WN, Denson LA, Baldassano RN, Hommel KA (2012) Treatment adherence in adolescents with inflammatory bowel disease: the collective impact of barriers to adherence and anxiety/depressive symptoms. J Pediatr Psychol 37:282–291
15. Modi AC, Quittner AL (2006) Barriers to treatment adherence for children with cystic fibrosis and asthma: what gets in the way? J Pediatr Psychol 31:846–858
16. Simons LE, McCormick ML, Devine K, Blount RL (2010) Medication barriers predict adolescent transplant recipients' adherence and clinical outcomes at 18-month follow-up. J Pediatr Psychol 35:1038–1048
17. Simons LE, Blount RL (2007) Identifying barriers to medication adherence in adolescent transplant recipients. J Pediatr Psychol 32: 831–844
18. Modi AC, Monahan S, Daniels D, Glauser TA (2010) Development and validation of the Pediatric Epilepsy Medication Self-Management Questionnaire. Epilepsy Behav 18:94–99
19. Solez K, Colvin RB, Racusen LC, Haas M, Sis B, Mengel M, Halloran PF, Baldwin W, Banfi G, Collins AB, Cosio F, David DS, Drachenberg C, Einecke G, Fogo AB, Gibson IW, Glotz D, Iskandar SS, Kraus E, Lerut E, Mannon RB, Mihatsch M, Nankivell BJ, Nickeleit V, Papadimitriou JC, Randhawa P, Regele H, Renaudin K, Roberts I, Seron D, Smith RN, Valente M (2008) Banff 07 classification of renal allograft pathology: updates and future directions. Am J Transplant 8:753–760
20. Haas M, Sis B, Racusen LC, Solez K, Glotz D, Colvin RB, Castro MC, David DS, David-Neto E, Bagnasco SM, Cendales LC, Cornell LD, Demetris AJ, Drachenberg CB, Farver CF, Farris AB 3rd, Gibson IW, Kraus E, Liapis H, Loupy A, Nickeleit V, Randhawa P, Rodriguez ER, Rush D, Smith RN, Tan CD, Wallace WD, Mengel M, Banff meeting report writing committee (2014) Banff 2013 meeting report: inclusion of c4d-negative antibody-mediated rejection and antibody-associated arterial lesions. Am J Transplant 14:272–283
21. Foster BJ, Dahhou M, Zhang X, Platt RW, Samuel SM, Hanley JA (2011) Association between age and graft failure rates in young kidney transplant recipients. Transplantation 92:1237–1243
22. Nascimento E, Fabreti de Oliveira RA, Maciel MD, Pereira AB, das Mercês de Lucas F Jr, Salomão-Filho A, Pereira WA, Moreira JB, Vilaça SS, de Castro Gontijo R, Lasmar MF, Vianna HR, Magalhães A, Calazans CA, Simão-Filho C, Vilela B (2014) Kidney transplantation: evaluation and clinical outcome of 237 recipients at low, medium, high, or strong immunological risk of rejection. Transplant Proc 46:101-107
23. Omoloja A, Mitsnefes M, Talley L, Benfield M, Neu A (2007) Racial differences in graft survival: a report from the North American Pediatric Renal Trials and Collaborative Studies (NAPRTCS). Clin J Am Soc Nephrol 2:524–528
24. Patzer RE, Mohan S, Kutner N, McClellan WM, Amaral S (2015) Racial and ethnic disparities in pediatric renal allograft survival in the United States. Kidney Int 87:584–592
25. Dunn TB, Noreen H, Gillingham K, Maurer D, Ozturk OG, Pruett TL, Bray RA, Gebel HM, Matas AJ (2011) Revisiting traditional risk factors for rejection and graft loss after kidney transplantation. Am J Transplant 11:2132–2143
26. Gonzales MM, Bentall A, Kremers WK, Stegall MD, Borrows R (2016) Predicting individual renal allograft outcomes using risk models with 1-year surveillance biopsy and alloantibody data. J Am Soc Nephrol 27:3165–3174
27. Eid L, Tuchman S, Moudgil A (2014) Late acute rejection: incidence, risk factors, and effect on graft survival and function. Pediatr Transplant 18:155–162
28. Favier LA, Taylor J, Loiselle Rich K, Jones KB, Vora SS, Harris JG, Gottlieb BS, Robbins L, Lai JT, Lee T, Kohlheim M, Gill J, Bouslaugh L, Young A, Griffin N, Morgan EM, Modi AC (2018) Barriers to adherence in juvenile idiopathic arthritis: a multicenter

collaborative experience and preliminary results. J Rheumatol 45: 690–696
29. Pascoe JE, Sawnani H, Hater B, Sketch M, Modi AC (2019) Understanding adherence to noninvasive ventilation in youth with Duchenne muscular dystrophy. Pediatr Pulmonol 54:2035–2043
30. Badawy SM, Thompson AA, Penedo FJ, Lai JS, Rychlik K, Liem RI (2017) Barriers to hydroxyurea adherence and health-related quality of life in adolescents and young adults with sickle cell disease. Eur J Haematol 98:608–614
31. Pollock-Barziv SM, Finkelstein Y, Manlhiot C, Dipchand AI, Hebert D, Ng VL, Solomon M, McCrindle BW, Grant D (2010) Variability in tacrolimus blood levels increases the risk of late rejection and graft loss after solid organ transplantation in older children. Pediatr Transplant 14:968–975
32. Goodall DL, Willicombe M, McLean AG, Taube D (2017) High intrapatient variability of tacrolimus levels and outpatient clinic nonattendance are associated with inferior outcomes in renal transplant patients. Transplant Direct 3:e192
33. Whitehouse W, Morris B (1997) Paediatric out-patient antiepileptic drug doses recorded in the medical charts are not reliable: implications for the notion of noncompliance. Seizure 6:41–42
34. Danziger-Isakov L, Frazier TW, Worley S, Williams N, Shellmer D, Dharnidharka VR, Gupta NA, Ikle D, Sweet SC, CTOTC-05 Consortium (2016) Perceived barriers to medication adherence in pediatric and adolescent solid organ transplantation. Pediatr Transplant 20:307–315
35. Danziger-Isakov L, Frazier TW, Worley S, Williams N, Shellmer D, Dharnidharka VR, Gupta NA, Ikle D, Shemesh E, Sweet SC, CTOTC-05 Consortium (2019) Perceived barriers to medication adherence remain stable following solid organ transplantation. Pediatr Transplant 23:e13361
36. Tadros HJ, Rawlinson AR, Martin E, Pietra BA, Fricker FJ, Gupta D (2020) Family functioning in pediatric heart transplantation: variables associated with poor outcomes. Pediatr Transplant 24: e13883
37. Viswanathan M, Golin CE, Jones CD, Ashok M, Blalock S, Wines RC, Coker-Schwimmer EJ, Grodensky CA, Rosen DL, Yuen A, Sista P, Lohr KN (2012) Closing the quality gap: revisiting the state of the science (vol. 4: medication adherence interventions: comparative effectiveness). Evid Rep Technol Assess (Full Rep). (208.4): 1–685
38. Wu YP, Pai ALH (2014) Health care provider-delivered adherence promotion interventions: a meta-analysis. Pediatrics 133:e1698–e1707
39. Kelly B, Squires JE, Feingold B, Hooper DK, Mazariegos GV (2019) Quality initiatives in pediatric transplantation. Curr Opin Organ Transplant 24:64–72

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